1) Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Assign static IP addresses on Windows

Perform these steps on each computer:

1. Open Control Panel → Network and Internet → Network and Sharing Center.
2. Select Change adapter settings.
3. Right-click the active Ethernet or Wi-Fi adapter and select Properties.
4. Select Internet Protocol Version 4 (TCP/IPv4), then click Properties.
5. Select Use the following IP address.
6. Enter the device's assigned IP address, subnet mask, and default gateway from the table.
7. Enter a DNS server if required, such as your router's IP address.
8. Click OK to save the settings.

2) Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

1. Create the network topology

Connect one server and three PCs to a switch.

DHCP Server

192.168.1.2

Switch

PC0

PC1

PC2

2. Configure the DHCP server

Click the server in Packet Tracer.

1. Open Desktop → IP Configuration.
2. Set the static configuration:

Setting	Value
IP address	192.168.1.2

| Subnet mask | 255.255.255.0 |
| Default gateway | 192.168.1.1 |

3. Open Services → DHCP.
4. Turn the DHCP service On.
5. Configure this DHCP pool:

Setting	Value
Pool name	LAB-POOL
Default gateway	192.168.1.1
DNS server	8.8.8.8
Start IP address	192.168.1.100
Subnet mask	255.255.255.0
Maximum users	50

6. Click Add or Save, depending on your Packet Tracer version.

**3) Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.**

roblem	**Possible cause**	**Solution**
nslookup times out	DNS service stopped or port 53 blocked	Check dnsmasq status and firewall
Hostname resolves to the wrong IP	Incorrect DNS record or another resolver is being used	Verify the record and client DNS server
DNS works but browser fails	Web server stopped or port 80 blocked	Check Nginx and HTTP connectivity
Works on the server but not other devices	Clients use a different DNS server	Configure DHCP/router DNS settings
.local behaves inconsistently	mDNS conflicts with unicast DNS	Use musicplayer.test for the lab

4) Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

1. Create the network topology

PC0

192.168.1.10/24

Switch

Router R1

LAN: 192.168.1.1

WAN: 203.0.113.2

External Network

203.0.113.0/24

Server: 203.0.113.10

The addresses in 203.0.113.0/24 are reserved for documentation. Use them only in the isolated simulation, not as real internet addresses.

2. Configure the router interfaces

Click the router and open the CLI. The following example uses a Cisco router with GigabitEthernet0/0 connected to the LAN and GigabitEthernet0/1 connected to the external network.

1) enable 2) configure terminal 3) interface gigabitEthernet0/0

4) ip address 192.168.1.1 255.255.255.0

5) ip nat inside

6) no shutdown

7) exit

1) interface gigabitEthernet0/1

2) ip address 203.0.113.2 255.255.255.0

3) ip nat outside

4) no shutdown

5) exit

If your router uses different interface names, use the available interfaces shown by show ip interface brief.

3. Configure NAT overload (PAT)

Enter the following commands:

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet0/1 overload

This configuration allows multiple private LAN devices to share the router's outside interface IP address using different port mappings.

Add a default route if your topology includes an upstream router:

ip route 0.0.0.0 0.0.0.0 203.0.113.1

Here, 203.0.113.1 represents the upstream router. Configure that device with a compatible address and a route back to the external server network as needed.

Save the configuration:

end

write memory

4. Configure the LAN PC

On PC0, open Desktop → IP Configuration and set:

Setting	**Value**
IP address	192.168.1.10
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server	Your configured DNS server

For this isolated lab, you can test by IP address without DNS.

5. Configure the external server

Add a server on the external network and configure:

- IP address: 203.0.113.10
- Subnet mask: 255.255.255.0
- Default gateway: 203.0.113.1

Enable its HTTP service under Services → HTTP.

Ensure the upstream router at 203.0.113.1 is configured and connected to both R1 and the external server's network, with appropriate routing. Without this upstream path, NAT alone cannot provide connectivity.

6. Test connectivity

From PC0, open Desktop → Command Prompt.